Artificial Intelligence for robotics

Part I : The different forms of AI

Simon Gay

Artificial Intelligence for Robotics

- **What is the objective of this course ?**
  - Learn the different forms of AI
  - Being able to implement core mechanism of different models of AI
  - Being able to use higher level tools (libraries, models)

- **What we won’t do ?**
  - Create a deep AI model from scratch
  - Create robots to dominate the world

Artificial Intelligence for Robotics

- **A fundamental question:**

What is Artificial Intelligence ?

  - **Let’s sort out fakes and facts...**

Artificial Intelligence for Robotics

- **A (even more) fundamental question:**

What is ~~Artificial~~ Intelligence ?

  - **And this is not that simple…**

  - *In 1986, more than twenty psychology experts were asked to provide a definition of intelligence, but there were no consensus (Sternberg, 1986)*

  - *The notion of intelligence can also differ according to culture !*

Artificial Intelligence for Robotics

- **Some capabilities seems widely accepted as indicators of intelligence:**
  - Learning abilities (Beckmann, 2006)
  - Abilities to adapt to the context (Piaget, 1953)
  - Ability to solve problems by using new solutions (Gardner, 1989)
- **Artificial Intelligence's objective is to reproduce such capabilities on artificial systems.**

Artificial Intelligence for Robotics

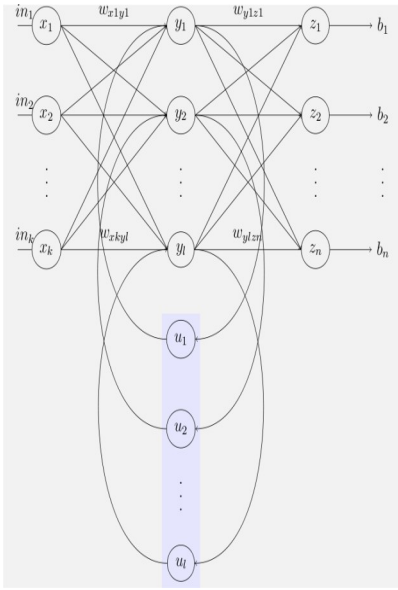
- **Since the beginning, two AI approaches emerged:**

  - **Symbolic AI:** algorithms based on symbols and logic rules. Solve through inference and deduction methods manipulating these rules and symbols.

  - **Connectionist AI:** algorithms based on sets of values and learning mechanisms enabling the convergence of these values to an optimal solution.

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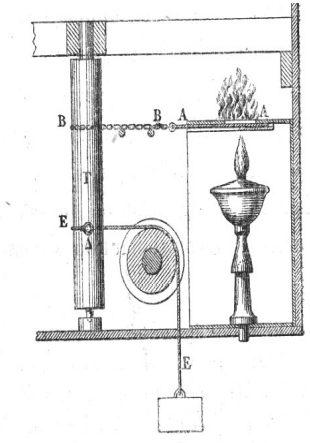
Artificial Intelligence for Robotics

- **And what about robotics in all this ?**
  - Robot (definition) : A mechanical device that is capable of performing a variety of often complex human tasks on command or by being programmed in advance.
- **A robot is a machine that perform tasks by interacting with a real-world environment**
  - **high complexity**
  - **changing**
- **Artificial intelligence would help adapting to the environment**

Artificial Intelligence for Robotics: History of AI

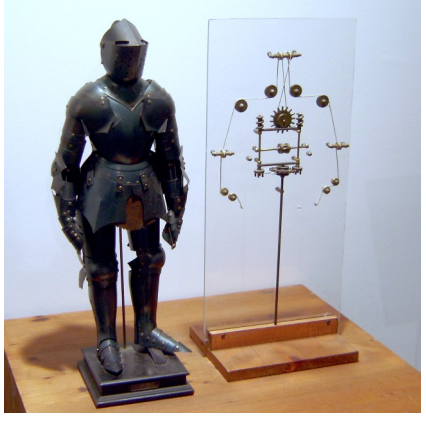
- **Ancient period: between myths...**
  - Vulcain's automatons, myth of Talos, Pygmalion's statue, Golem.

  ... and first automatons
  - Egypt: temple animated statues
  - Greece: theater automatons, Antikythera mechanism (mechanical astronomic computer)
- **Middle Age and Renaissance: automatons become more complex**
  - Mechanical knight of Léonard de Vinci
  - Vaucanson's automatons (XVIIIth)
  - The first robot ? Tea serving automaton (XVIIth)









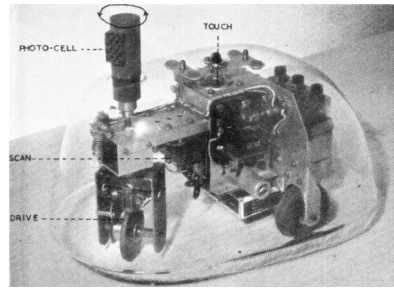




Artificial Intelligence for Robotics: History of AI

- **XXth century: emergence of electronics**
  - Electric dog by John Hammond and Benjamin Miessner (1912)
  - Karel Čapek's piece Rossum's Universal Robots (1921): invention of term **'robot'**
  - emergence of *cybernetics* in 1942
    - Autonomous Tortoises of Walter Grey (1950)
  - First forms of Machine Learning (1951): logistic regression, K-nearest neighbors





- **1956: Artificial Intelligence officially becomes a scientific research domain (Dartmouth conference)**

  Excitement and optimism: every laboratory wants a AI department

  - Perceptrons: first artificial neural network (1957)
  - Logic programming (symbolic IA, 1958)
  - Evolutionist approaches (1965)
  - K-means (1967)

http://cyberneticzoo.com/cybernetic-time-line/

Artificial Intelligence for Robotics: History of AI

- **1974: first crash of AI**
  - Due to the lack of results, funding is cut everywhere in the world → first AI winter

- **Beginning of 80s: renewed interest with Expert Systems**
  - *Total* domination of symbolic approaches
  - Robot Shakey: symbolic reasoning algorithms, A*
    - But required simplified and mostly static environment
  - Other approaches emerge (but nobody care) : Reinforcement learning (1988), multi-agent systems (middle of 80s), subsumption architecture (1986), multi-layer neural network, auto-adaptative maps

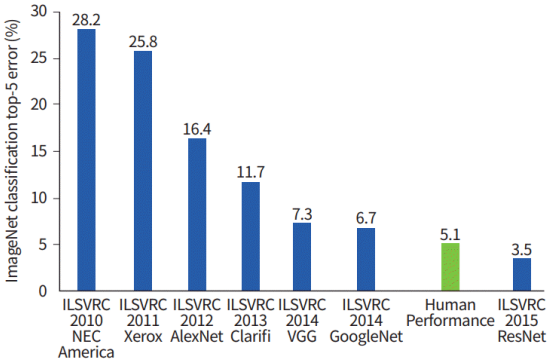
- **Middle of 90s: lack of progress in Expert Systems**
  - Second winter of IA

Artificial Intelligence for Robotics: History of AI

- **1997: Deep Blue computer (IBM) beats chess world champion Kasparov**
  - Re-renewed interest for IA both in research and industry
    → second gold age of IA
  - Emergence of developmental robotics (middle 90s) then developmental intelligence (middle 2000s)
  - Emergence, around 2010, of networks with high number of layers (but nobody believe in it)



- **2012: a Deep Learning model (AlexNet) beats all records in image recognition contest ILSVRC**
  - Deep learning becomes an active research domain
    - 2014: generative adversarial networks
    - 2017: text analysis (Transformer)
    - 2020: text generation and LLM (GPT-3)
    - 2022: text-to-image (Stable Diffusion)
    - 2023: multi-modal networks (text, image, videos)
  - Total reign of deep neural networks



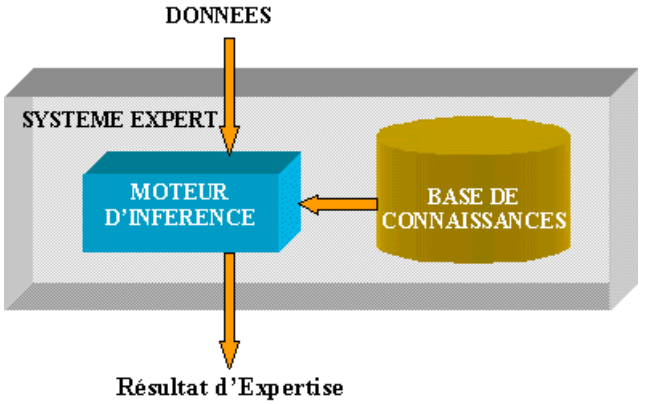
Artificial Intelligence for Robotics: forms of AI

Multiple types of AI

/!\ There is no official classification

- **Symbolic: inference and deductions based on symbols and rules**
  - Logic Programming
  - Expert Systems
- **Connectionism: adjusting numerical values and connections between components**
  - Neural Networks
  - Logistic regression
  - Reinforcement Learning
  - ….
- **Emergence: solving problems through interactions of simple structures**
  - Systèmes multi-agents
  - Subsumption Architecture
- **Evolutionary: natural selection inspired models**
  - Genetic algorithms

Artificial Intelligence for Robotics: forms of AI



- **Symbolic AI**
  - Set of rules provided by the conceptor
  - Based on the manipulation of 'symbols'
  - Most known model: Expert Systems
    - Try to reproduce the knowledge of an exprt human in a specific domain
  - Logic programming language:
    - PROLOG
  - No learning in most cases

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- **Applications**
  - Widely used in the field of decision-making within a specific area (ex : medicine)

    advantages: simple and light softwares, make possible to explain results

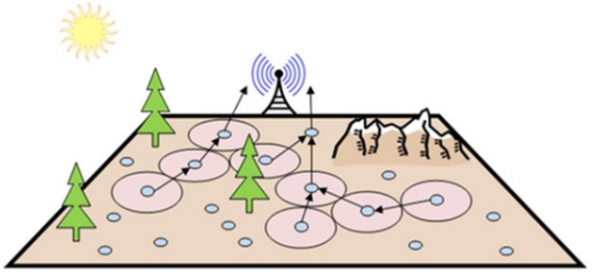
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Artificial Intelligence for Robotics: forms of AI

Emergence models

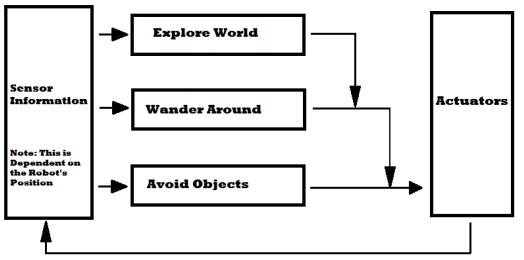


- **Multi-agent Systems**
  - Partially bio-inspired: social insects (ants, bees…), bird, fish
  - Systems made of very simple elements interacting with each other and with the environment
  - Emergence of a global behavior (‘the whole is greater than the sum of its parts’)
  - Do not require learning



Applications

- Simulation of complex phenomena
- Routing optimization
- Cinema and video games (large armies)



- **Subsumption Architecture**
  - Multiple reactive behavior modules
  - Hierarchy between modules
  - Complex behaviors emerge to respond to the environment

Applications



- Domestic robots (vacuum robots, lawnmower robots)
- Exploration robots (mars rovers, undersea exploration robots)

Artificial Intelligence for Robotics: forms of AI

Multiple types of AI

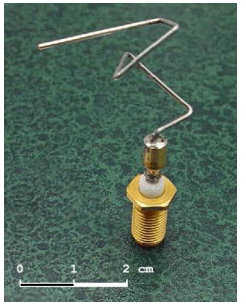
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Artificial Intelligence for Robotics: forms of AI

- **A particular type of learning model: genetic algorithms**
  - Inspired by species evolution and natural selection
  - “learning” (optimization) at each new generation

  - **Principle:**
    - First population with random parameters (genome)
    - Simulation and performance measure, best genomes are kept and combined (with random mutations) to generate next generation.
    - Optimization of parameters at each new generation

  - **Applications:**
    - Optimization, routing
    - Automatic adjustment of robots’ controllers



Spacecraft antenna

Artificial Intelligence for Robotics: forms of AI

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Artificial Intelligence for Robotics: forms of AI

- **Connectionnist IA: a whole family with multiple learning approaches**

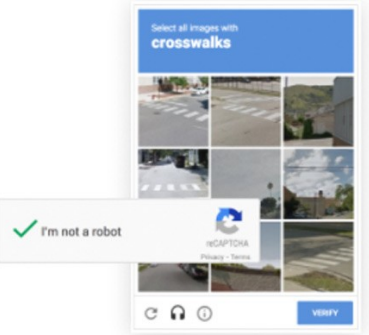
  - **Supervised learning:** models that first learn with a database of input data with associated expected outpout (training dataset). The model is then exploited on data with unknown output.

  - **Non-supervised Learning:** models that do not required datasets with known output. The system is trained to converge toward an optimal solution.

  - **Reinforcement learning:** the system can act and interact with its environment and is involved in its learning process. The system learns consequences and changes produced by its actions.

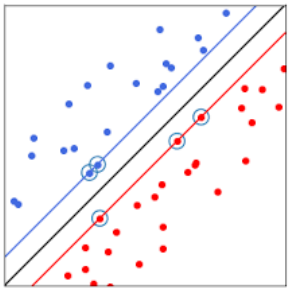
Artificial Intelligence for Robotics: forms of AI

- **Supervised learning**



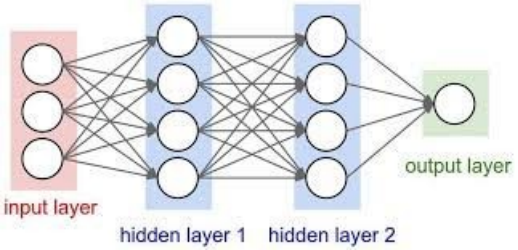
  - The model is trained on a dataset with known results.

    → A dataset with hand-labelled result must be first generated



  - Mathematical models:
    - K-NN, Logistic Regression, SVM...

  - Biologically-inspired approaches: the neural networks
    - Composed of formal neurons (extremely simplified models of neurons).
    - A whole family of existing networks
    - Work like a classifier
    - Very good at detecting statistical regularities, but also bias



  - **Application :**
    - Any application requiring classification: image recognition, text analysis, translation, decision making…
    - … as soon as you have a sufficient training dataset

Artificial Intelligence for Robotics: forms of AI

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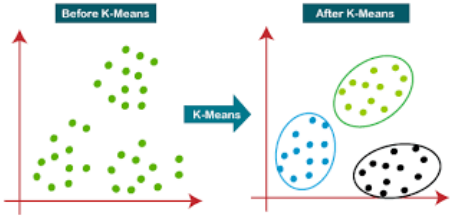
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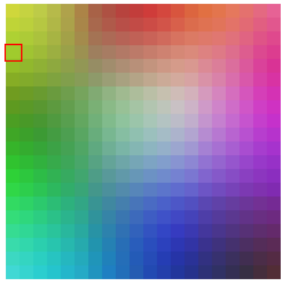
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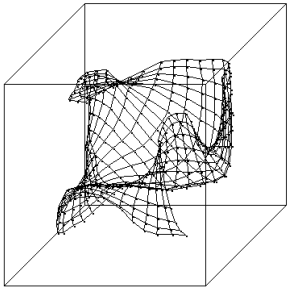
Artificial Intelligence for Robotics: forms of AI

- **Non-supervised learning**
  - Learning process without known results
    - → approaches used more to organize information
    - → Convergence of parameters toward an optimal solution (at least locally)
  - Mathematical approaches:
    - K-means, hierarchical clustering



  - Biologically-inspired approaches: non supervised neural networks
    - Associative memory (Hopfield Networks)
      - Recursive and asynchronous networks able to store images
    - Self Organizing Maps (SOM)
      - Inspired by brain's cortices
      - Can map large dimension data on 2D matrix while keeping proximity between input data elements





  - **Applications :**
    - Brain studies, statistics, data analysis

Artificial Intelligence for Robotics: forms of AI

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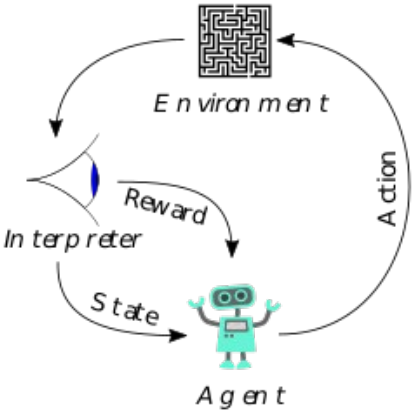
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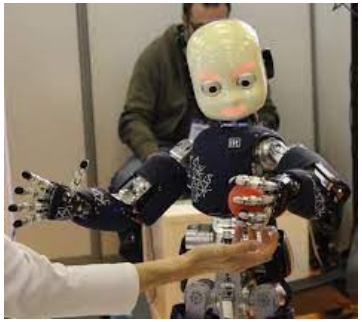
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Artificial Intelligence for Robotics: forms of AI

- **Connectionnist IA: reinforcement learning**
  - Embodied Intelligence: an artificial agent interact with an environment (real or virtual). The agent perceives its environment, acts and learns consequences of its actions
  - Goal-oriented approaches: a reward model is defined on achieving a given task or goal
    - Reinforcement Learning (1988): the agent learns environment state changes and generate behaviors maximizing the reward
    - Q-Learning (1989): similar but focuses on actions allowing to reach rewards
    - **Applications:** robot control, trading, virtual bots...



  - Behavior-based approaches: research domains on behavior emergence and sensorimotor learning in human and animals, no rewards but intrinsic motivation, no states but sensorimotor learning, bio-inspired learning models
    - Developmental robotics (middle 90s): reproducing development stages in infants or animals (walk, stand, eye-hand coordination...)
    - Developmental intelligence (middle 2000s): studying emergence of behaviors and environment understanding in simple creatures only through sensorimotor learning
    - **Applications:** researches in cognition, neurology, psychology...



Artificial Intelligence for Robotics: summary

